Prática de Física dos Dispositivos Eletrônicos

FGA0100

Laboratório-3

Termistores

01/2022

FGA

Universidade de Brasília

Resistivities and Temperature Coefficients of Resistivity for Various Materials

Material	Resistivity[a] ($\Omega \cdot m$)	Temperature Coefficient α [$(°C)^{-1}$]
Silver	1.59×10^{-8}	3.8×10^{-3}
Copper	1.7×10^{-8}	3.9×10^{-3}
Gold	2.44×10^{-8}	3.4×10^{-3}
Aluminum	2.82×10^{-8}	3.9×10^{-3}
Tungsten	5.6×10^{-8}	4.5×10^{-3}
Iron	10×10^{-8}	5.0×10^{-3}
Platinum	11×10^{-8}	3.92×10^{-3}
Lead	22×10^{-8}	3.9×10^{-3}
Nichrome[b]	1.50×10^{-6}	0.4×10^{-3}
Carbon	3.5×10^{-5}	-0.5×10^{-3}
Germanium	0.46	-48×10^{-3}
Silicon	640	-75×10^{-3}
Glass	10^{10} to 10^{14}	
Hard rubber	$\sim 10^{13}$	
Sulfur	10^{15}	
Quartz (fused)	75×10^{16}	

[a]All values are at 20°C.

[b]Nichrome is a nickel–chromium alloy commonly used in heating elements.

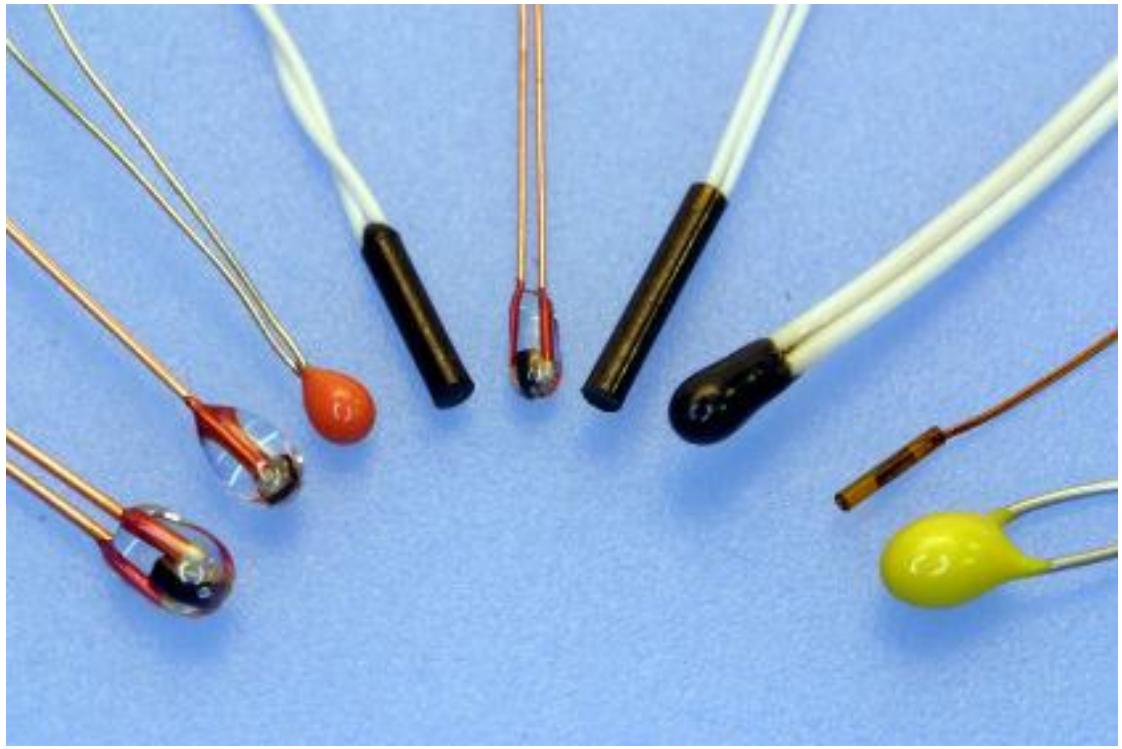
Termistores PTC

Termistores PTC são feitos de Materiais Cerâmicos ou Compósitos Plásticos

Cerâmica Policristalina Dopada contendo Titanato de Bário ($BaTiO_3$),
ou ainda variações com misturas de Titanatos de Bário, Chumbo, Estrôncio
com a adição de Ítrio, Manganês, Tântalo e Sílica.

Compósito Plástico contendo Partículas ou Grãos de Carbono ou de Metais.

Termistores NTC

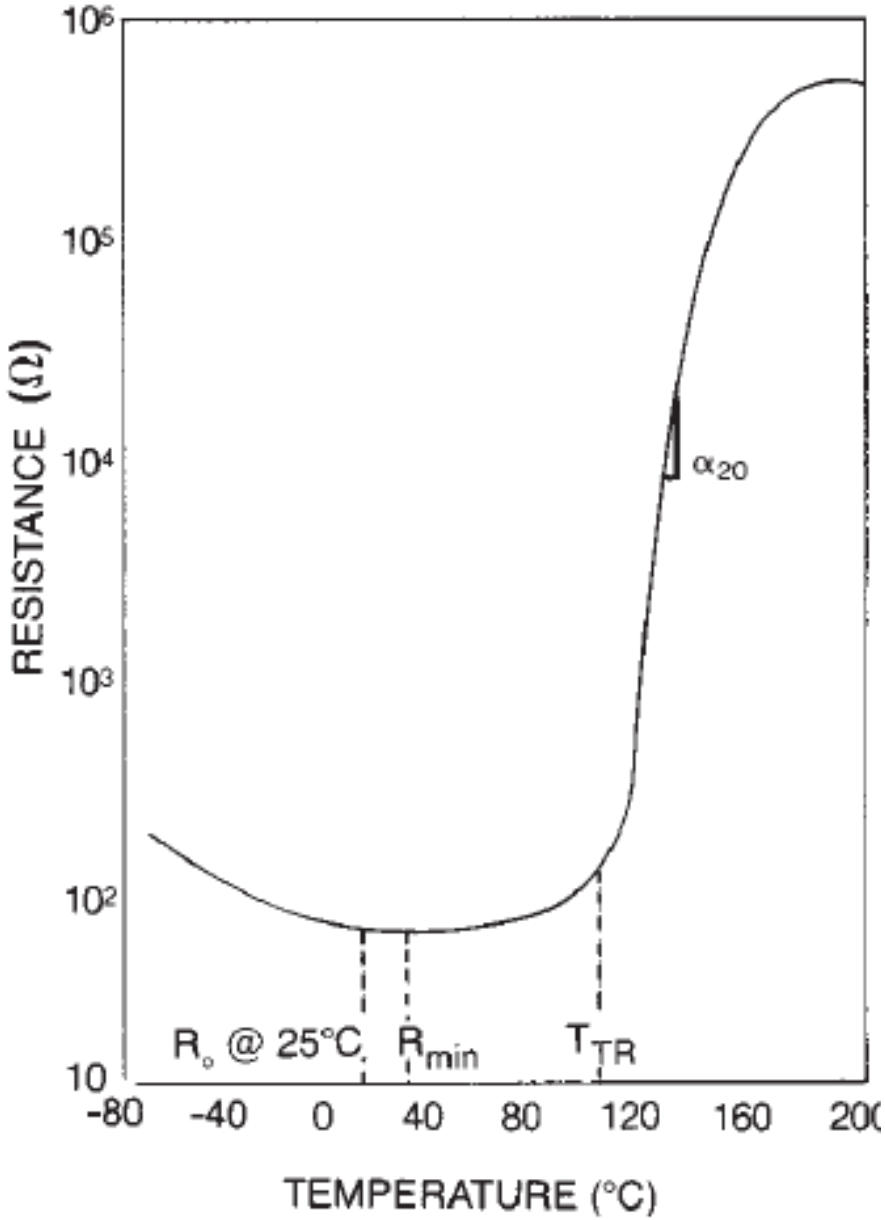


Termistores NTC feitos de Óxidos Metálicos Semicondutores Sinterizados e Dopados.

Óxido Férrico (Fe_2O_3) dopado com Titânio (Ti) [tipo-*n*]

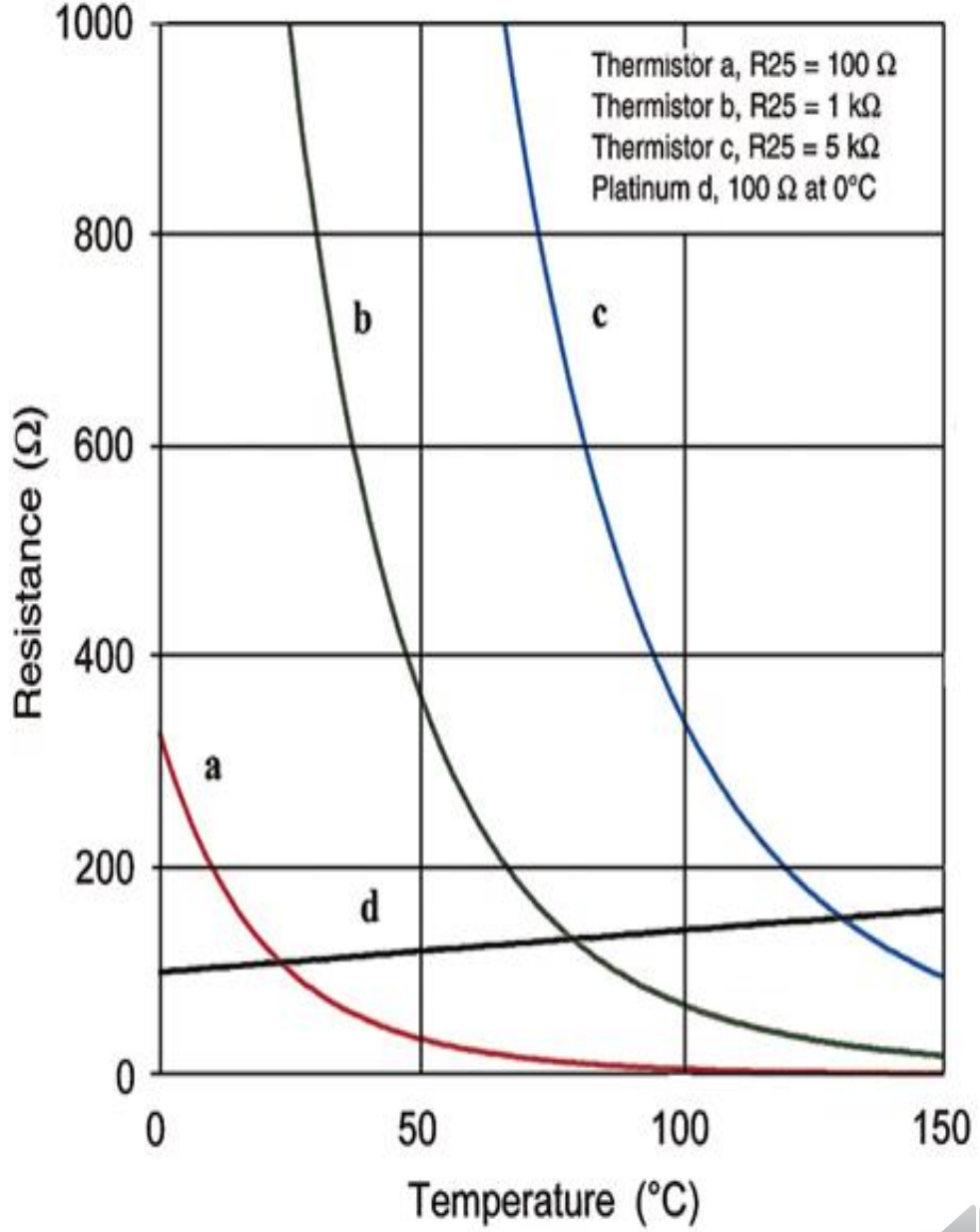
Óxido de Nickel (NiO) dopado com Lítio (Li) [tipo-*p*]

PTC



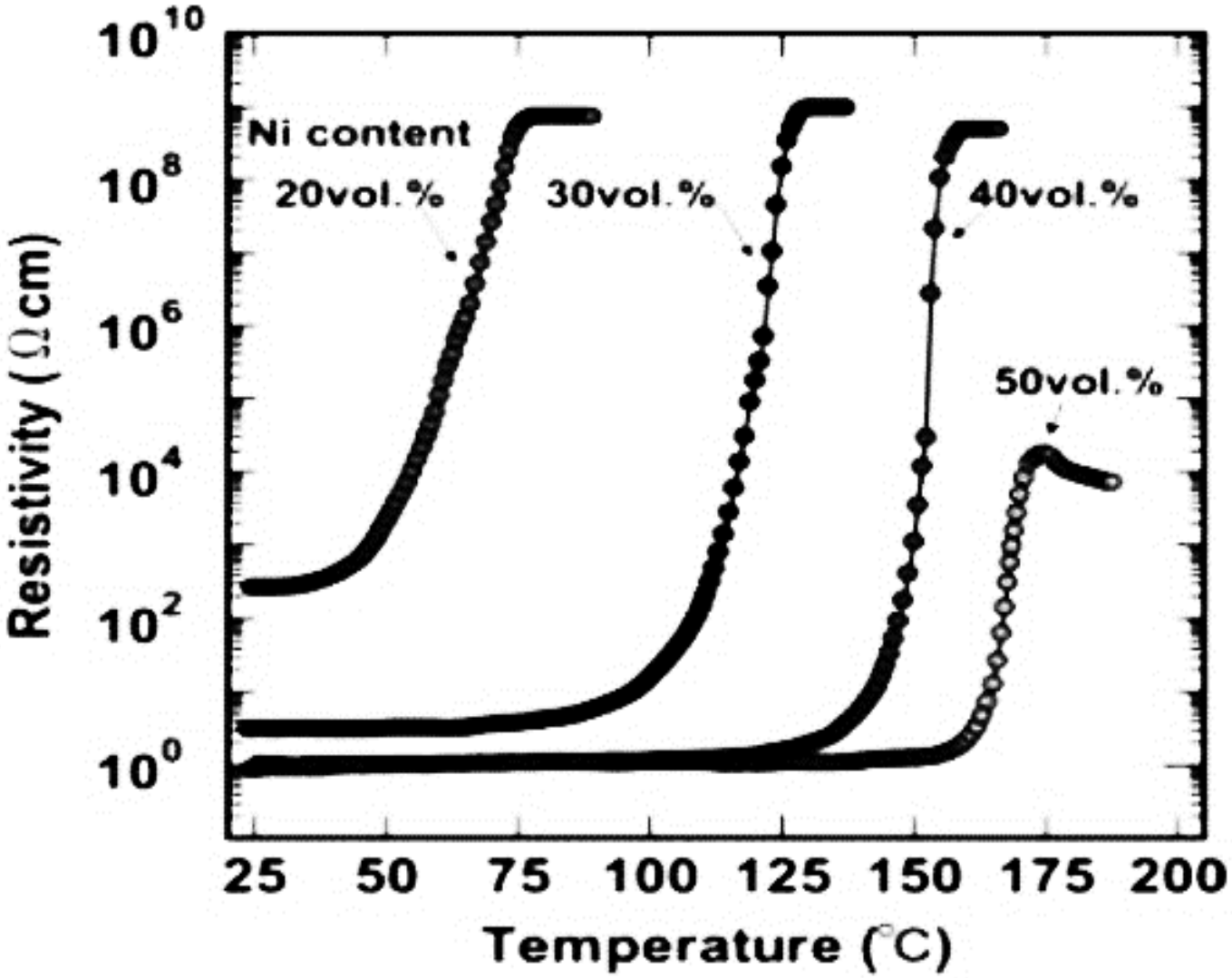
Cerâmico (Titanato de Bário)

NTC



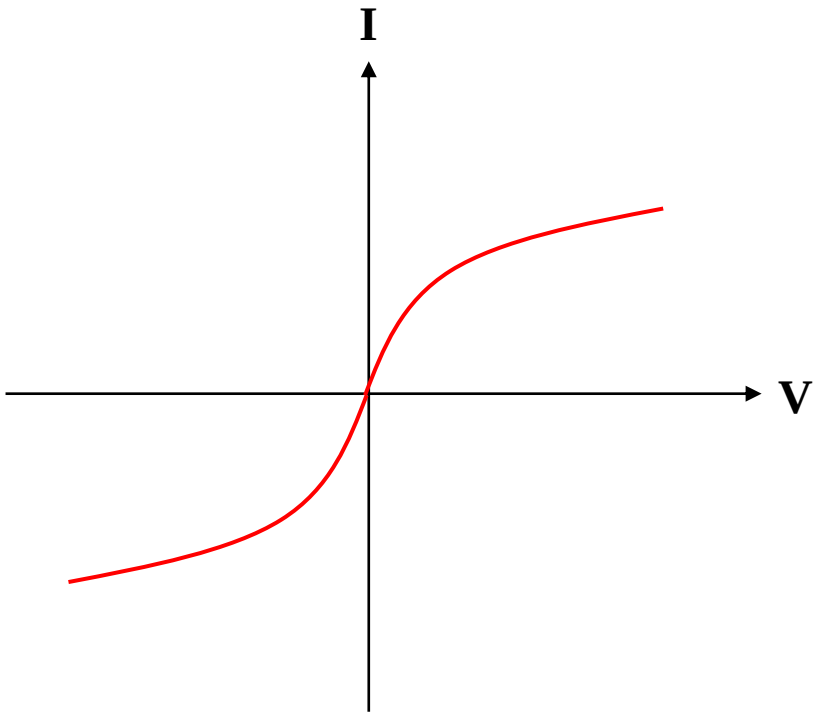
Fe_2O_3 Dopado

Termistores PTC de Compósitos Plásticos



Grãos de Níquel em Fluoreto de Polivinilideno

Termistores



PTC

$$G = \frac{dI}{dV}$$

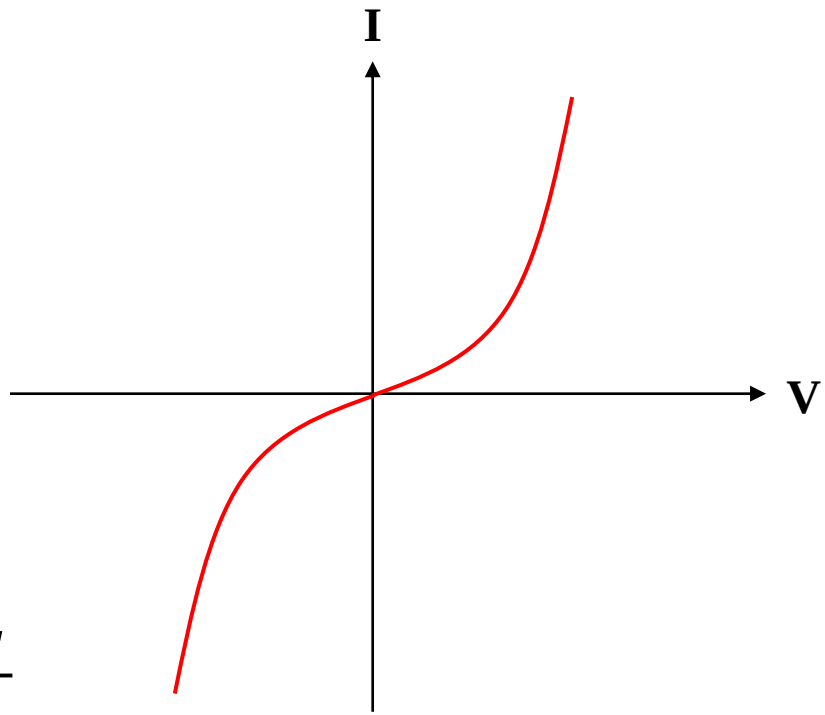
$$R = \frac{dV}{dI}$$

$$G = \frac{1}{R}$$

$$R = \rho(T)\frac{L}{A}$$

$$G = \sigma(T)\frac{A}{L}$$

$$\sigma(T) = \frac{1}{\rho(T)}$$



NTC

Modelo para o Termistor

Convecção → $P_T = K\left[T(R) - T_0\right]$

Radiação

Condução

$$P_E = IV$$

Equilíbrio Térmico
(alto-aquecimento)

$$P_E = P_T$$

$$I^2 R = K\left[T(R) - T_0\right]$$

$$\boxed{T_0 = T(R) - \frac{I^2 R}{K}}$$

(ar parado)

Coeficiente Térmico

$$\alpha_T = \frac{1}{R(T)} \left.\frac{dR}{dT}\right|_T$$

Modelo do NTC: Equação de Steinhart-Hart

$$\frac{1}{T} = \frac{1}{T_0} + \frac{1}{B}\ln\left(\frac{R}{R_0}\right)$$

T_0 (25 °C = 298.15 K)

$$\boxed{R = R_0 e^{-B\left(\frac{1}{T_0} - \frac{1}{T}\right)}}$$

$$R = r_\infty e^{B/T} \qquad r_\infty = R_0 e^{-B/T_0}$$

Modelo do NTC: Equação de Steinhart-Hart

Obtendo as Temperaturas das Medidas:

Datasheet do NTC 5D-9:

$$r_\infty = 0{,}18\Omega \qquad R_0 = 5{,}0\Omega \qquad T_0 = 298{,}15K$$

(25 ºC)

(Corrente máxima de 3A)

Obter o coeficiente B:

$$B = T_0 \log\left(\frac{R_0}{r_\infty}\right) \text{ [K]}$$

Obter a Temperatura:

$$T = \frac{B}{\log\left(\frac{R_1}{r_\infty}\right)} \text{ [K]}$$

Valores Calculados de R_1 a partir das medidas.

Modelo do NTC: Equação de Steinhart-Hart

Obtendo a constante K de Convecção:

$$T_0 = T(R) - \frac{I^2 R}{K} = T(R) - \frac{P_T}{K} \qquad P_E = P_T$$

$$K = \frac{P_T}{T(R) - T_0}$$

	Type Number	Zero Power Resistance At 25°C	Max.Steady State Current At 25°C	Thermal Dissipation Constant	Thermal Time Constant	Operating Temperature Range
		Ω	A	mW/°C	Sec	°C
NTC	5D-9	5	3	11	34	-55 ~ +200

Modelo Alfa de Termistor PTC

$$\alpha = \frac{\ln\left(\frac{R_2}{R_1}\right)}{T_2 - T_1}$$

Onde:

R_1 = Resistência em T_1 [Ohms]

R_2 = Resistência em T_2 [Ohms]

T_1 = Temperatura-1 [K]

T_2 = Temperatura-2 [K]

Modelo do PTC

$$\alpha = \frac{\ln\left(\frac{R_2}{R_1}\right)}{T_2 - T_1} \quad \Rightarrow \quad R_2 = R_1 \cdot e^{\alpha \cdot (T_2 - T_1)}$$

$$R(T) = R_0 e^{\alpha(T - T_0)}$$

$$T = \frac{1}{\alpha} ln\left[\frac{R(T)}{R_0}\right] + T_0$$

Válido para a região abrupta da curva.

Modelo Beta de Termistor NTC

$$\beta = \frac{\ln\left(\frac{R_{T1}}{R_{T2}}\right)}{\left(\frac{1}{T_1} - \frac{1}{T_2}\right)}$$

Onde:

R_{T1} = Resistência em T_1 [Ohms]

R_{T2} = Resistência em T_2 [Ohms]

T_1 = Temperatura-1 [K]

T_2= Temperatura-2 [K]

Modelo Dinâmico dos Termistores

$$P = V \times I = \frac{dH}{dt} = K[T(R) - T_0] + C_T \frac{dT(R)}{dt}$$

C_T – Capacidade Térmica [Joule / K]
H – Calor [Joules]
P – Potência Elétrica
V – Tensão [volts]
I – Corrente [amps]
K – Constante de Dissipação de Calor
por Convecção Natural [W / K]

Modelo Dinâmico do Termistor NTC

$$P = V \times I = \frac{dH}{dt} = \underbrace{K[T(R) - T_0]}_{\text{Dissipação Térmica (Convecção)}} + \underbrace{C_T \frac{dT(R)}{dt}}_{\text{Inércia Térmica}}$$

$$P(t) = V^2(t)/R(t) \qquad I(t) = V\,(t)/R(t)$$

$$R = R_0 e^{-B\left(\frac{1}{T_0} - \frac{1}{T}\right)}$$

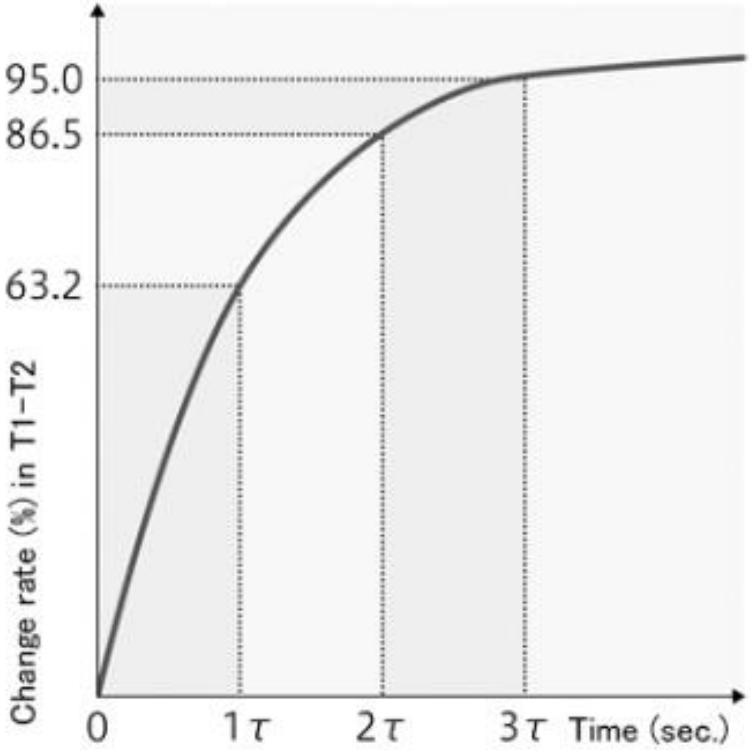
Thermal Time Constant (TTC)

$$T = (T_2 - T_1)\left(1 - \exp(-t/\tau)\right) + T_1$$

$$T = (T_2 - T_1)\left(1 - e^{-1}\right) + T_1$$

$$\frac{T - T_1}{T_2 - T_1} = 1 - e^{-1} = 1 - \frac{1}{2.718} = 0.632$$

63.2%



Thermal Time Constant (TTC)

	Type Number	Zero Power Resistance At 25°C	Max.Steady State Current At 25°C	Thermal Dissipation Constant	Thermal Time Constant	Operating Temperature Range
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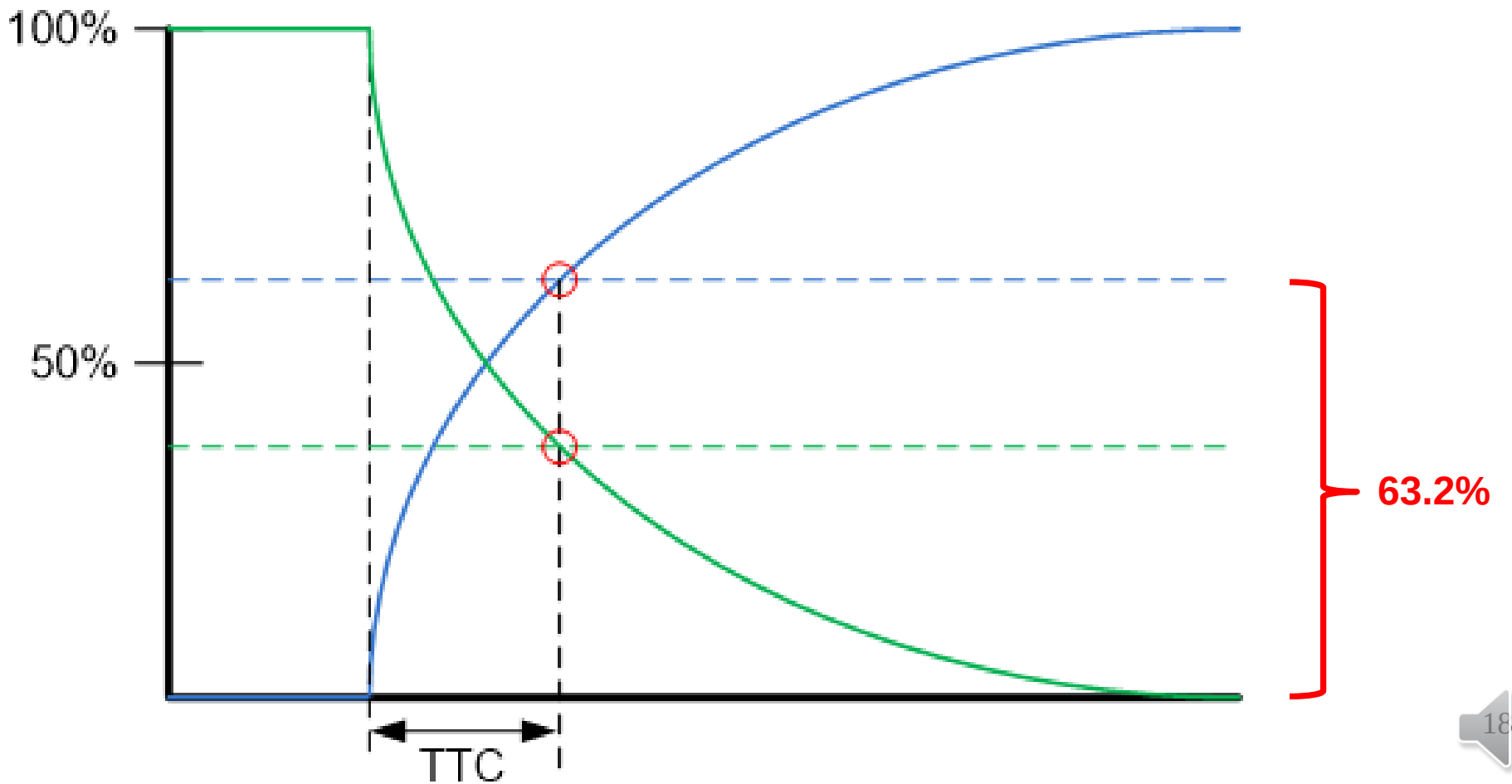
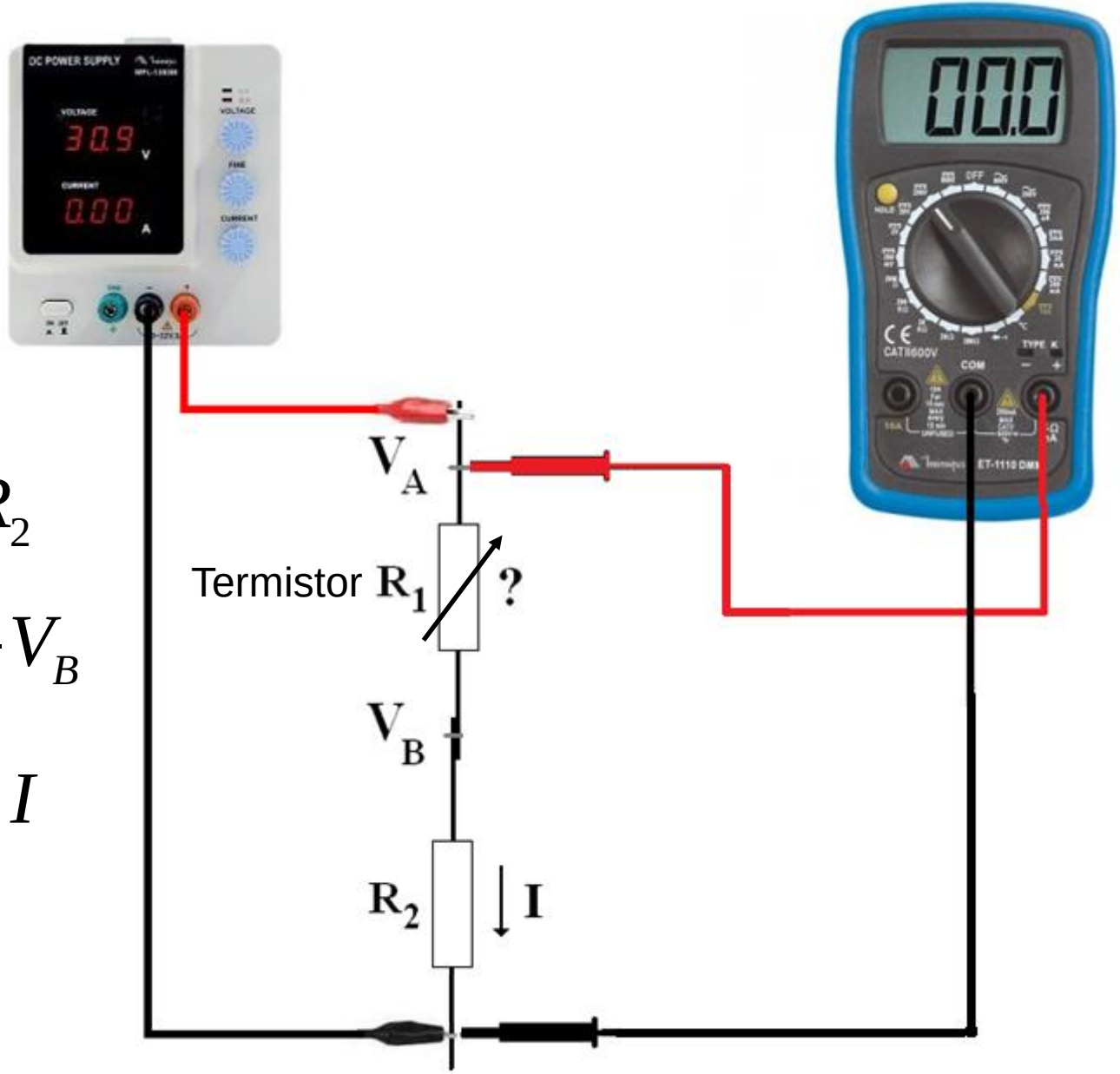


Circuito de Medidas

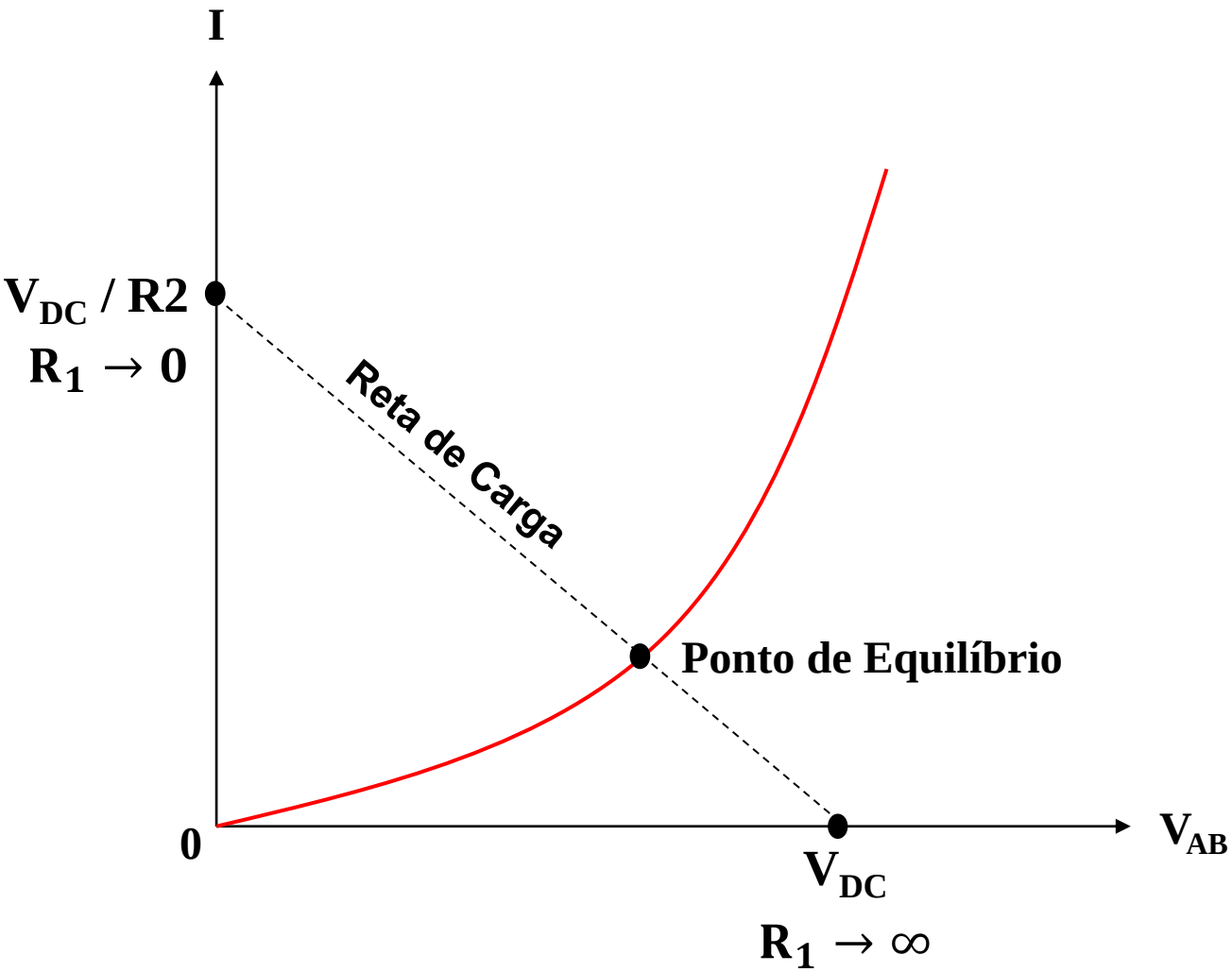
$$I = V_B / R_2$$

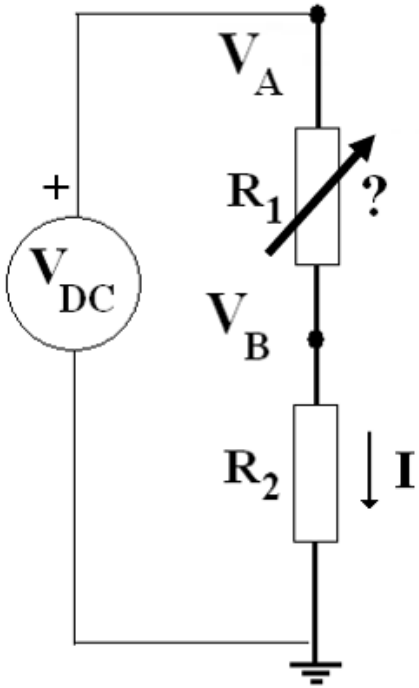
$$V_{AB} = V_A - V_B$$

$$R_1 = V_{AB} / I$$

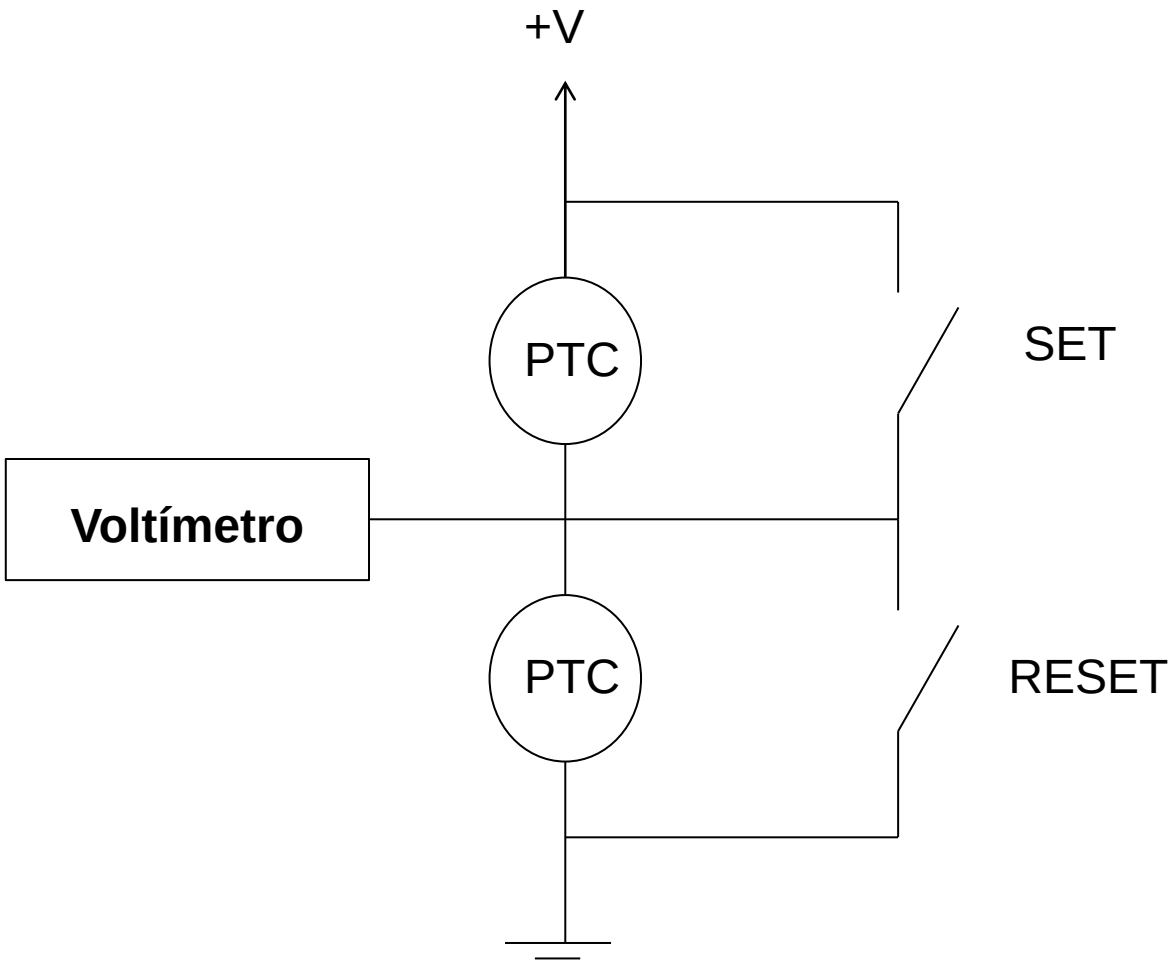


Termistor NTC

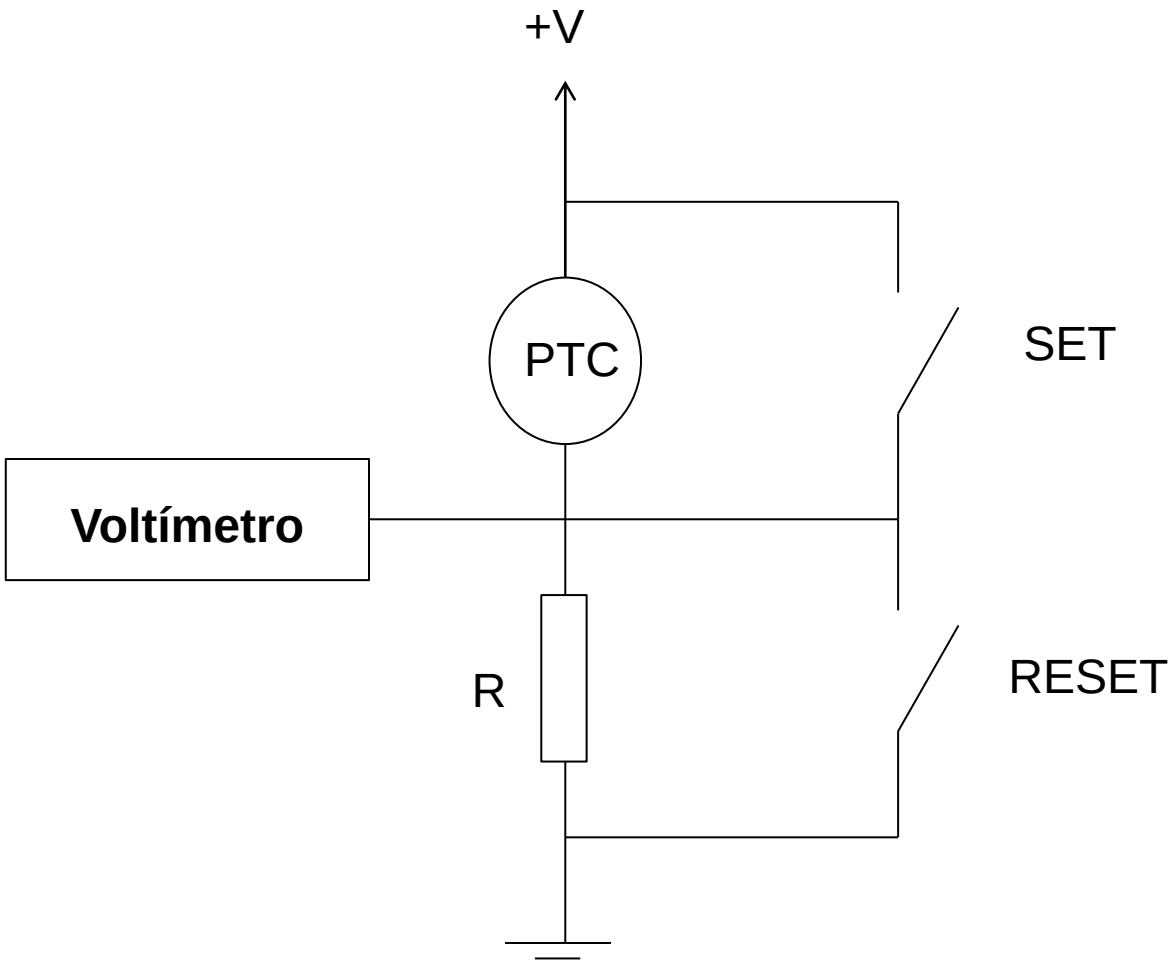




Efeito Memória em Circuitos com Termistores



Efeito Memória em Circuitos com Termistores



Aplicação Tecnológica: Termistor PTC como Medidor de Fluxo

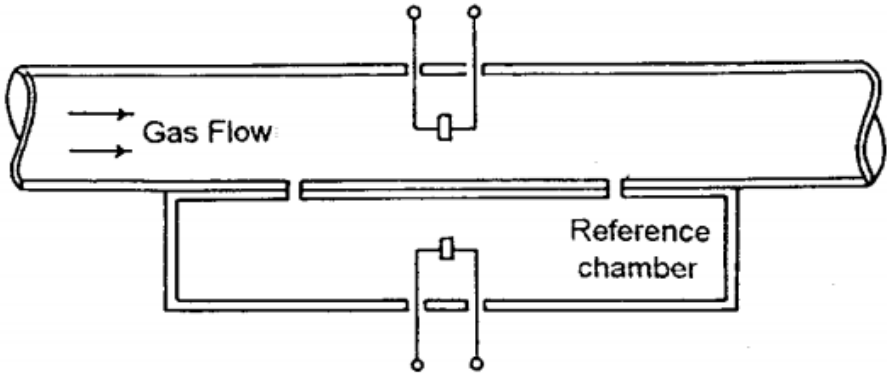


FIG. 2. Schematic structure of a PTC based flow sensing head.

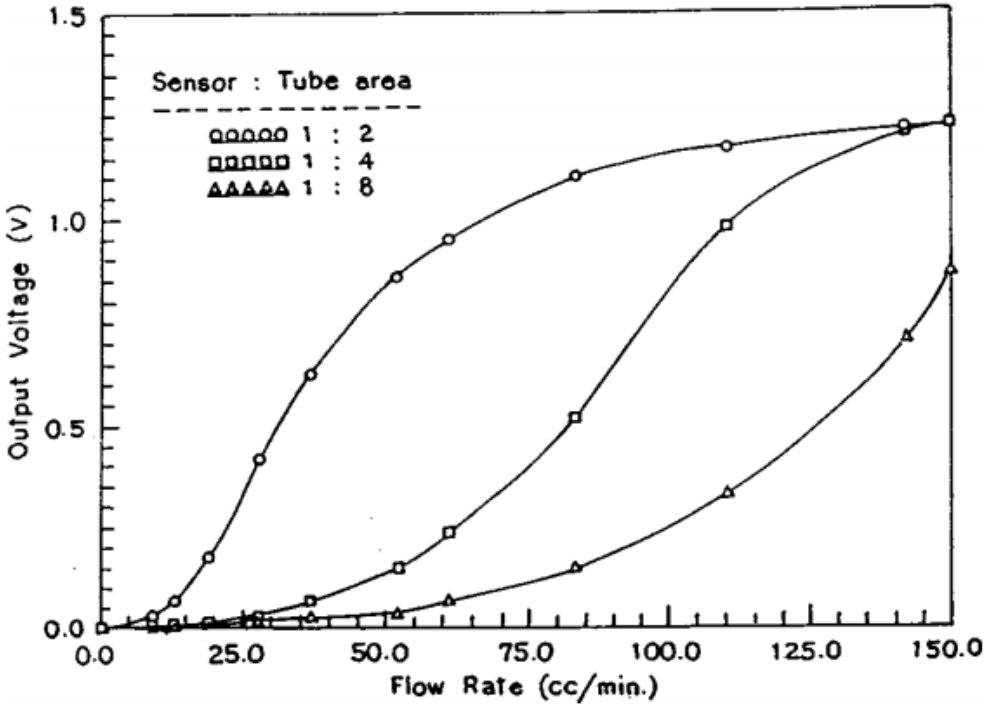


FIG. 8. Output voltage response of a PTC flow sensor with varying cross section of the tube.

https://www.researchgate.net/publication/234921674_Gas_flow_meter_using_a_positive_temperature_coefficient_thermistor_as_the_sensor